

Experiment-15

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Course Code: -MLA0305

Title: Policy Optimization Using Simulated Experiences

Aim:

To implement model-based policy optimization using simulated experiences and compare its sample efficiency with model-free Reinforcement Learning.

Procedure:

1. Select a suitable Reinforcement Learning environment.
2. Collect real environment experiences.
3. Train an approximate environment model.
4. Generate additional simulated experiences using the model.
5. Use the simulated experiences to train the policy.
6. Train a model-free RL agent using only real experiences.
7. Record the number of environment interactions required.
8. Compare cumulative rewards.
9. Analyze sample efficiency.
10. Plot real samples versus performance.



The screenshot shows a Jupyter Notebook window titled 'Untitled16.ipynb'. The interface includes a top menu bar with 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help'. Below the menu is a toolbar with 'Commands', '+ Code', '+ Text', and 'Run all'. On the right side of the toolbar, there are icons for 'Share', 'RAM', and 'Disk'. The left sidebar contains navigation icons for the notebook's structure. The main area displays a code cell with the following Python code:

```
import numpy as np
policy=np.array([.5,.5]); predicted=np.array([4.,8.])
for i in range(10):
    z=predicted-predicted.max(); p=np.exp(z); p/=p.sum(); policy=.7*policy+.3*p; print(i+1,np.round(policy,3))
```

The output of the code cell is a list of 10 rows, each containing two values rounded to three decimal places:

```
1 [0.355 0.645]
2 [0.254 0.746]
3 [0.183 0.817]
4 [0.134 0.866]
5 [0.099 0.901]
6 [0.075 0.925]
7 [0.058 0.942]
8 [0.046 0.954]
9 [0.037 0.963]
10 [0.032 0.968]
```

Result:

The model-based method used simulated experiences to improve learning efficiency. Its sample efficiency was compared with the model-free approach.